

Satyam Kumar

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EDUCATION

NYU Courant Institute of Mathematical Sciences

Master of Science in Computer Science

Birla Institute of Technology and Science Pilani

Bachelor of Engineering in Electronics and Communication

New York

August 2024 – May 2026

Hyderabad, India

August 2018 – May 2022

EXPERIENCE

Research Assistant, Prof. Sherry Yang and Prof. Yann LeCun (CILVR Lab, NYU) July 2025 – August 2026

- **Diffusion Forcing Transformer for Video Generation** Optimized an autoregressive **Diffusion Forcing transformer** framework by replacing standard encoder with domain-adapted, fine-tuned VAE. Engineered a hybrid training objective combining **JEPA temporal prediction** with **I-JEPA / MAE-style spatial masking** to **improve latent dynamics**, resulting in improved **FVD and FID scores** for continuous gameplay simulation.
- Used **WorldGym** to evaluate the Video generation model by rolling a policy out inside an action conditioned video world model from just a single start frame and using vision-language reward model to estimate success rate and compare policies.
- Built Push-Letters (similar to **pushT**); collected expert dataset; integrated with **DINO-WM** pipeline to validate **latent-space planning and Model predictive control (MPC)** for tasks. Validated test-time planning with embeddings from learned world-model (no rewards or demonstrations), showing robust zero-shot goal reaching on the new shape set and supporting the paper's task-agnostic results across manipulation suites.

Data Science and Machine Learning Intern, Curve Biosciences

June 2025 – Dec 2025

- Built downstream **Multiple Instance Learning (MIL) Framework** on top of the team's **genomic foundation model** embeddings, designing **Perceiver** and **Hierarchical Attention Transformer** architectures for early-stage chronic disease detection. Collaborated with researchers to integrate machine learning insights into ongoing biomedical studies, contributing to the development of computational tools for detection of early stage chronic diseases.
- Developed and optimized **PyTorch-based implementations** with **DDP distributed training** across multi-GPU nodes, focusing on scalability, modularity, and experimental reproducibility. Engineered a **GCP-backed data pipeline** with prefetching and caching to **eliminate data-loading bottlenecks** on large-scale datasets, profiling to identify where GPU utilization was being lost. Benchmarked deep learning approaches against **applied statistical baselines (Regression, Decision Trees)** for early cancer diagnosis.

Member of Technical Staff, EdgeVerve Systems

July 2022 – July 2024

- **Multi-Modal Question Answering System:** Worked on extracting information from videos, images, and PDF documents leveraging computer vision, audio processing, OCR, and Natural Language Processing techniques.
- **LLM Agents:** Developed single and multiple LLM agents using **LangChain** and **LangGraph** for **video summarization and question answering**. **Text-to-SQL:** Implemented data retrieval from SQLite database using **ReAct (Reasoning and Acting)** in LLMs by translating natural language into SQL queries.
- Enhanced the development, optimization, and deployment of various product components, including model trainers, model managers, and prediction APIs (**REST API**) for computer vision tasks. Utilized **PostgreSQL** for robust data storage, **Redis** for caching mechanisms, and **RabbitMQ** for message brokering.

PROJECTS

RL Post-Training for Math Reasoning: SFT + GRPO

Feb 2026 – March 2026

- Implemented SFT and GRPO from scratch (no TRL/Trainer) to post-train Qwen2.5-Math-1.5B, with vLLM rollout generation in a two-GPU policy/inference setup. SFT on chain-of-thought traces raised MATH-500 accuracy from 60.8% to 65.7%; GRPO with verified rewards raised Countdown dev accuracy from 15.3% to 76.5% over 500 rollout steps.
- Ran ablations on advantage normalization (Dr. GRPO), length normalization, baselines, and learning rate, logging gradient norm, clip fraction, and token entropy in W&B to diagnose training stability.

OpenAI's Model Craft: Parameter Golf Challenge

March 2026 – April 2026

- Engineering a compact language model constrained to 16 MB (weights + code) trained in under 10 minutes on 8×H100 GPUs, evaluated by tokenizer-agnostic bits-per-byte on FineWeb. Proposed a parameter-efficient architecture combining depth recurrence with aggressive quantization, an enriched subword tokenizer, and online test-time adaptation via low-rank updates jointly optimizing model capacity, compression ratio, and inference-time adaptability within the fixed byte budget.

ARC AGI Challenge, Prof. Saining Xie

March 2025 – May 2025

- Engineered a 14B LLM using LoRA-based cognitive priors, custom grid tokenization, and a dual-stage transduction-induction solver (DeepSeek-R1 base) for complex reasoning tasks. Achieved 29% accuracy on the challenging ARC benchmark by integrating test-time adaptation and 4-way ensemble voting while also identifying key model failure modes.

TECHNICAL SKILLS

Languages & Courses: Python, C/C++, Machine Learning, Large Lang. Vision Models, GPU, RL for Foundational models

Frameworks & Libraries: PyTorch, Tensorflow, JAX, CUDA, HuggingFace, Flask, FastAPI, scikit-learn, LangChain

Tools and Infra: GCP, AWS, Linux, Ray, MCP, Cursor, Copilot, Docker, Kubernetes, Weights & Biases, MuJoCo

Research Areas : Video Generation, World Models, Diffusion models, Training and fine-tuning of Foundational models, RL post-training